

What the Weekend Clock Does to Option Prices

A companion note: the weekend variance discount, translated into premium

19 August 2026

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A companion note to The Price of Calendar Time in a Market That Never Closes. Nothing here is used in that paper; it translates its results into premium. Reproducible from `scripts/weekend_price_impact.py` and `scripts/weekend_price_observed.py`; tables land in `output/tables/p*.csv` and `o*.csv`, the figure in `output/figures/p_f1_price_impact.*`.

The question

The paper is written in variance. It says that Deribit quotes weekend calendar time at roughly half to two thirds of the weekday rate, and argues about whether that ratio matches the one realized returns actually deliver. Variance is the right object for the economics and the wrong one for anyone who pays for options. The natural follow-up is the mechanical one: **volatility is lower over a weekend, so by Black-Scholes the price must be lower — by how much?**

The direction is right and the size is not obvious, because three things sit between a variance ratio and a premium, and they push in different directions.

1. **Dilution.** The ratio applies to weekend *hours*, not to contracts. What reaches a price is the ratio weighted by the share of that contract's remaining life falling on Saturday and Sunday. Across every short-dated trade in the sample that share averages 0.216, so the typical contract sees about a fifth of the raw effect. A quarterly option is 2/7 weekend whatever day it is quoted on, and the clock washes out of it entirely.
2. **The square root.** Price responds to volatility; the clock is stated in variance. Half the effect is gone before anything else happens.
3. **Convexity.** Price is close to linear in volatility at the money and is not linear at all away from it. The elasticity $\sigma * \text{vega} / C$ is 1.22 in the at-the-money bucket and 4.30 in the far wing,

so the same proportional cut in volatility takes three and a half times as large a share of a wing premium. And the wings are where the paper's §7 finds the market applying a *deeper* clock to begin with, so the two amplifications compound.

The short answer: the weekend clock removes **7.0% of the time value** of the average short-dated Bitcoin option, **4.9% at the money and 17.2% in the far wing**; on a Friday daily expiring Monday it removes **13.0% at the money and 62% two standard deviations out**. Over the sample that is **\$910m of premium across the four books**, of which about **\$117m** is more than realized weekend variance justifies — a figure that has changed sign since 2022 and is not statistically distinguishable from zero book by book.

1. The arithmetic

Write w for the fraction of a contract's remaining life falling on a weekend and R for the weekend variance ratio the market quotes. Total variance to expiry is

$$V = v_{wd} \cdot T \cdot (1 - (1 - R) \cdot w)$$

so the contract's implied volatility is the weekday rate scaled by the square root of an **effective-time factor** $f = 1 - (1 - R)w$. That factor is the whole story: it is the ratio of effective time to calendar time, and it is what the option is really being sold.

Everything below compares one contract against **itself with the clock switched off**: same strike, same expiry, same instant, same forward, volatility raised from the quoted level to that level divided by the square root of f . That comparison is deliberate. It is exactly the within-instant contrast the headline regression identifies, and because both legs are built from the same contract-level weekday variance, the level of the surface — and with it whatever variance risk premium the market is charging — cancels out of every number reported here.

Two conventions worth stating. Percentages are quoted on *time value*, that is, on the out-of-the-money leg at each strike. By put-call parity the forward-intrinsic part of an in-the-money premium carries no volatility, so the dollar effect of a change in the clock is identical whichever leg you price, while the percentage is not: dividing a real dollar effect by a premium that is nine tenths intrinsic reports a small number about a large one. Second, premia are reconstructed from the exchange's own implied volatility through Black-76 rather than read off the traded price. Section 7 checks that reconstruction against the premia that were actually paid and against volatilities inverted from them, and nothing in the note turns on it.

In volatility points

The blunt version, at each book's fitted weekday rate:

book	weekday IV	Friday daily → Monday ($w = \frac{2}{3}$)	all-weekend contract ($w = 1$)
BTC	66.5	57.8 (−8.6 pts)	53.0 (−13.5 pts)
ETH	83.3	72.8 (−10.5 pts)	66.9 (−16.4 pts)
SOL	89.9	75.5 (−14.4 pts)	67.2 (−22.7 pts)
XRP	103.7	84.2 (−19.6 pts)	72.4 (−31.3 pts)

A market maker who ignored the weekend and quoted one flat surface would be offering Friday's three-day Bitcoin contract nearly nine volatility points too rich, and XRP's nearly twenty.

In premium

p1_price_impact_stylized.csv, Bitcoin, three days to expiry, at the full-sample quoted clock R = 0.635:

weekend fraction	effective time	IV	ATM	1 sd out	2 sd out
0 (Tue → Fri)	3.00 d	66.5	—	—	—
2/7 (long-dated)	2.69 d	62.9	-5.4%	-15.2%	-30.5%
1/2	2.45 d	60.1	-9.6%	-26.4%	-49.7%
2/3 (Fri → Mon)	2.27 d	57.8	-13.0%	-35.1%	-62.4%
1 (Saturday daily)	1.91 d	53.0	-20.3%	-52.1%	-81.7%

Concretely: Bitcoin at \$100,000, a three-day at-the-money call listed Friday at 08:00 and expiring Monday at 08:00 is worth **\$2,090**. The identical contract with no weekend in it is worth **\$2,403**. The weekend takes \$313, or 13%, off the price — and takes 62% off the same contract struck two standard deviations away.

That last row is worth pausing on, because it is where the naive version of the question breaks down. “Vega times the change in volatility” would price that 2-sd Saturday contract at **-129%** of its own premium. The exact repricing is **-81.7%**. Convexity is not a correction here; it is most of the answer, which is why every number in this note is an exact repricing rather than a greek multiplication.

2. On the tape

The stylized map assumes a contract. p2-p3 reprice every trade that clears the headline filter — 13.6m Bitcoin trades, 9.2m Ether, 0.42m Solana, 0.14m XRP, all with 0.25 to 14 days to run — at that trade’s own quoted volatility, its own strike, its own maturity and its own weekend fraction. Averages are weighted by the counterfactual premium, so a book is described by the money that changes hands in it rather than by its cheapest contracts, and the weight does not respond to the effect being measured.

Pooled (p6):

book	mean w	IV cut	elasticity	premium cut	time value traded	premium removed
BTC	0.216	-4.10%	1.77	-7.03%	\$8,178m	\$618m
ETH	0.216	-3.98%	1.77	-6.86%	\$3,836m	\$283m
SOL	0.218	-5.05%	1.70	-8.29%	\$67m	\$6.0m
XRP	0.215	-5.86%	1.64	-9.36%	\$25m	\$2.6m

The dilution step dominates: a raw variance ratio of 0.635 becomes a 4.1% volatility cut once weighted by how much weekend the traded book actually carries. Convexity then puts roughly three quarters of it back, and the premium effect lands at 7%. The ordering across books follows the quoted ratios directly — XRP discounts the weekend hardest and pays for it with the largest premium effect.

By distance from the money (p2, Bitcoin):

bucket	mean w	IV cut	elasticity	premium cut	time value
at the money	0.212	-4.02%	1.22	-4.86%	\$4,455m
near	0.216	-4.09%	1.91	-7.57%	\$2,451m

bucket	mean w	IV cut	elasticity	premium cut	time value
wing	0.223	-4.23%	2.86	-11.29%	\$921m
far wing	0.242	-4.62%	4.30	-17.22%	\$347m

The volatility cut is nearly flat across the smile — this column holds the clock at the pooled estimate — and the premium cut rises three and a half fold. That is convexity alone, before any of §7's wing result is allowed in. The other three books give the same picture: far-wing premium cuts of 16.4% (ETH), 19.4% (SOL) and 23.1% (XRP).

By maturity (p3): the effect is essentially flat in maturity within the sample — Bitcoin runs -6.9%, -7.3%, -5.6%, -7.6% across the sub-daily, 1-3 day, 3-7 day and 7-14 day bands — because what varies is not the clock but the weekend content of each band, and the 3-7 day band happens to average the least weekend ($w = 0.171$) of the four. The sample stops at 14 days; past a few weeks the weekend fraction converges on 2/7 for every contract and the cross-sectional effect disappears by construction.

3. The wings pay for it twice

§7 finds that the far wings apply a *deeper* weekend clock than the money — Bitcoin's far-wing implied ratio is 0.528 against 0.630 at the money — and reads that as the smile's moneyiness metric following the clock about a third of the way rather than as a view on weekend tails. In premium terms that structural finding compounds with the convexity above. Repricing each bucket against the ratio the market actually quotes there (p5, Bitcoin):

bucket	IV cut	elasticity	premium cut	gap vs realized
at the money	-4.09%	1.22	-4.93%	+0.68%
near	-4.05%	1.91	-7.49%	+1.28%
wing	-4.59%	2.85	-12.13%	+0.89%
far wing	-6.52%	4.22	-22.65%	-3.12%

Nearly a quarter of a far-wing Bitcoin option's time value is the weekend clock. And the sign of the pricing gap is not the same across the smile: the money is quoted slightly rich against realized weekend variance while the far wing is quoted **cheap** by 3.1% — Bitcoin's far wing over-discounts the weekend enough to cross the realized level even though the book as a whole does not. Ether is the same shape (-3.5% in the far wing, +0.48% at the money); Solana and XRP, which over-discount everywhere, reach -10.4% and -13.7% in their far wings.

4. What of it is wrong

Setting the clock to each asset's *realized* weekend ratio instead of its quoted one prices the paper's headline gap in premium. This is the number a trader would care about, and it is small:

book	quoted R	realized R	gap in premium	dollars
BTC	0.635	0.584	+1.09%	+\$88m
ETH	0.645	0.607	+0.82%	+\$31m
SOL	0.558	0.657	-2.02%	-\$1.4m
XRP	0.488	0.621	-2.68%	-\$0.7m

Positive means the weekend-heavy contract is rich relative to a weekday contract quoted at the same instant. Bitcoin and Ether under-discount and their weekend contracts are about one percent of time value too expensive; Solana and XRP over-discount and theirs are two to three percent too cheap.

Two cautions, both inherited from the paper and neither optional. First, none of these four gaps is individually significant (§5.1): the variance ratios carry standard errors that straddle the realized values, so the dollar figures above are point estimates of quantities that cannot be signed asset by asset. Second, the comparison is to *realized* variance, and §7 spends a whole section establishing that no non-negative price of jump risk closes it — which is what licenses calling it a gap rather than a premium. Absent that section these numbers would be a description, not an error.

5. It has roughly tripled, and it has changed sign

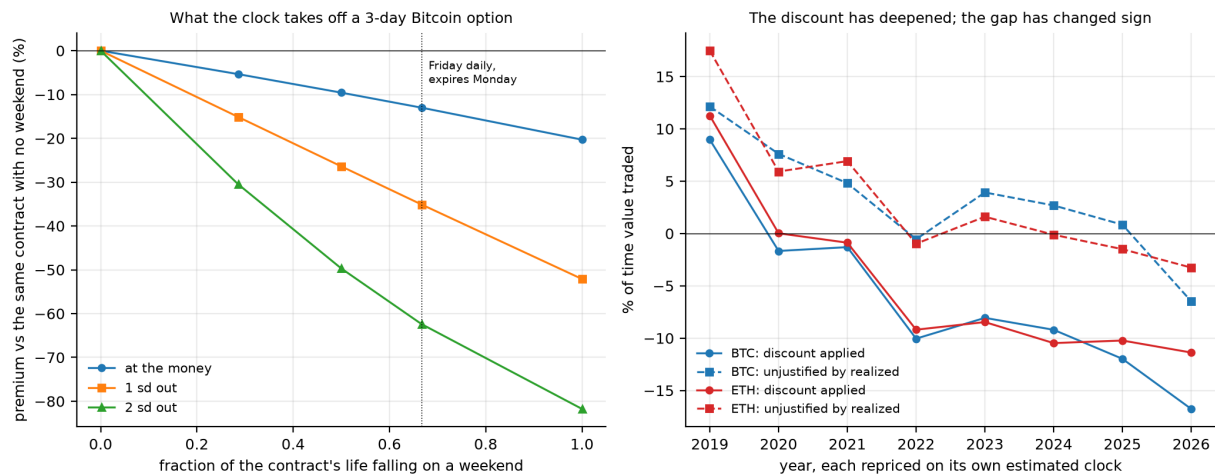
The tables above hold the clock at its full-sample value, so year-to-year movement in them is contract mix and nothing else. §5.5 measures the clock *itself* deepening by about 0.14 a year in Bitcoin and 0.12 in Ether. Repricing each year's traded book against that year's own estimated ratio (p8) is where the price story actually lives:

year	BTC quoted R	premium cut	gap	premium removed	gap \$
2019	1.57	+8.99%	+12.2%	-\$6m	+\$8m
2020	0.906	-1.67%	+7.60%	\$5m	+\$23m
2021	0.931	-1.30%	+4.80%	\$23m	+\$82m
2022	0.477	-10.03%	-0.54%	\$93m	-\$5m
2023	0.580	-8.05%	+3.92%	\$58m	+\$25m
2024	0.506	-9.19%	+2.68%	\$187m	+\$48m
2025	0.424	-11.97%	+0.85%	\$257m	+\$16m
2026*	0.296	-16.77%	-6.48%	\$153m	-\$53m

*2026 runs to mid-August.

Through 2021 the market applied essentially no weekend discount: Bitcoin's estimated ratio runs 1.57, 0.91, 0.93 across 2019–21, so weekend-heavy contracts were quoted *richer* than weekday ones in 2019 and within a percent and a half of them in 2020 and 2021. By 2026 the weekend clock is removing a sixth of the time value of every short-dated Bitcoin option traded. Ether tracks it: +11.3% in 2019, -11.4% in 2026.

The gap column is the part that matters for whether anyone is wrong. It starts at +12% of time value — the market charging for weekend variance that never arrived, which is the mispricing §6's calendar spread was harvesting — falls through zero around 2022, and is now **negative in both mature books**. The trade that made money for most of this sample is, on 2026 quotes, pointing the other way. That is the same fact §6 records as decaying profitability, seen in premium rather than in Sharpe ratios, and it is the reason the paper's open items lead with *why the market is still deepening a discount it has already overshot*.



The weekend clock in premium. Left: what a three-day Bitcoin option loses as its life fills with weekend, at three distances from the money, on the full-sample quoted clock. Right: each year's traded book repriced against that year's own estimated clock — the solid lines are the discount the market applies, the dashed lines the part of it realized weekend variance does not justify.

6. What a position earns across a weekend

Everything above is a cross-section: what a contract is quoted at relative to another contract at the same instant. A position also moves *through* the weekend, and that is a different number. Three calendar days pass from Friday 08:00 to Monday 08:00 and only about two effective days do, so a long option decays more slowly than the calendar says (p4, Bitcoin, at the money):

maturity on Friday	decay, flat calendar	decay, weekend clock	premium saved
4 days	-50.0%	-44.7%	+5.3 pts
5 days	-36.7%	-31.6%	+5.2 pts
7 days	-24.4%	-20.1%	+4.3 pts
10 days	-16.3%	-14.3%	+2.0 pts
14 days	-11.3%	-9.5%	+1.9 pts
30 days	-5.1%	-4.4%	+0.8 pts
60 days	-2.5%	-2.1%	+0.4 pts

A long weekly straddle carried over the weekend keeps four percentage points of premium a flat-calendar theta model would have written off, and the effect dies away with maturity because a long contract's weekend fraction barely moves over one weekend. The mirror image is the Friday mark-down: a seven-day Bitcoin contract quoted Friday morning already carries a 5.4% volatility mark-down for the weekend ahead, which is what the buyer is paying to avoid and the seller has already given up.

These two views are consistent and easy to confuse. The weekend is cheap to buy *and* cheap to hold; what it is not is free, and the cross-sectional discount is precisely the compensation for the slower decay.

7. The same numbers, on premia that were actually paid

Everything above is built from premia reconstructed through Black-76 from the exchange's implied volatility. The fair objection is that a pricing effect stated in reconstructed prices is partly a state-

ment about the pricer. The tape carries the traded premium itself, so the objection can be answered rather than conceded. Four checks, in increasing order of how little each assumes (o1-o5, from `scripts/weekend_price_observed.py`).

The reconstruction is a change of units. Every trade's observed premium against the Black-76 price at the exchange's own volatility, judged on time value:

book	trades	median error	within 1%	within 5%	$\Sigma \text{ recon} \div \Sigma \text{ observed}$	at the tick
BTC	13,590,752	-0.02%	43.0%	90.9%	0.9983	17.7%
ETH	9,217,106	-0.03%	47.9%	92.4%	0.9996	13.6%
SOL	422,919	-0.00%	96.0%	99.5%	1.0000	0%
XRP	135,445	-0.00%	96.7%	99.5%	1.0000	0%

The aggregates — which is what the dollar tables rest on — are right to within two parts in a thousand. The per-trade dispersion on the inverse books is the premium tick: Bitcoin quotes in increments of 0.0005 BTC, so a cheap option's price is coarsely quantized and no continuous formula can match it within one percent. That shows up as a wide error distribution and no bias. Fewer than one trade in a thousand prints below forward intrinsic; they are counted, not dropped.

The dollar figures, restated on observed money. Time value is directly observable — premium paid less forward intrinsic, both known per trade — so the base can be actual money with no pricer in it. Only the counterfactual leg stays modelled, and it has to: the contract with the weekend taken out of it does not exist to be observed.

book	time value traded, observed	reconstructed	premium removed, observed	reconstructed	gap, observed
BTC	\$8,194m	\$8,178m	\$620m	\$618m	+\$88.3m
ETH	\$3,839m	\$3,836m	\$283m	\$283m	+\$31.2m
SOL	\$66.8m	\$66.8m	\$6.03m	\$6.03m	-\$1.37m
XRP	\$25.2m	\$25.2m	\$2.59m	\$2.59m	-\$0.69m

Nothing moves. The pooled Bitcoin premium cut is -7.032% on observed time value against -7.029% reconstructed.

The effect itself, with no pricer in the outcome. Regressing log observed time value on the weekend fraction inside a cell of trades that printed on the same day, at the same strike to within two percent, in the same maturity decile, measures the premium discount straight off traded prices. Running the identical regression on the reconstructed series and estimating the difference directly — by regressing the log ratio of the two, so the difference carries its own day-clustered standard error — is the test:

book	bucket	observed	reconstructed	difference	s.e.
BTC	all	-0.6560	-0.6537	-0.0023	0.0024
BTC	at the money	-0.5476	-0.5457	-0.0019	0.0012
BTC	far wing	-0.5444	-0.5434	-0.0010	0.0072
ETH	all	-0.5683	-0.5682	-0.0001	0.0010
SOL	all	-0.4833	-0.4854	+0.0021	0.0027
XRP	all	-0.4895	-0.4927	+0.0032	0.0038

Across all sixty rows of o4 — three specifications, five buckets, four books — the largest divergence anywhere is 0.005 on a coefficient of -0.42, and only three rows reach two standard errors. Traded prices and reconstructed prices carry the same weekend effect.

Volatilities inverted from the traded premia. On a subsample stratified by year, Brent inversion of the observed price, owing the exchange’s volatility field nothing:

book	n	exchange IV	inverted IV	median gap	premium cut, exchange IV	inverted
BTC	39,975	77.22	77.41	+0.001 pts	-6.879%	-6.874%
ETH	31,990	82.35	82.40	+0.000 pts	-6.723%	-6.715%
SOL	12,000	83.45	83.45	+0.000 pts	-8.150%	-8.150%
XRP	12,000	93.30	93.31	+0.001 pts	-9.020%	-9.021%

The whole calculation run on volatilities recovered from traded prices lands within a hundredth of a percentage point of the same calculation on the exchange’s field.

One thing this exercise does *not* license. `o4` also carries the local derivative the pricer implies, $-\text{elasticity} \cdot (1-R)/2f$, and in the wings it sits well away from the measured coefficient — Bitcoin’s far wing is -0.54 measured against -1.22 predicted. That gap is the estimator’s, not the market’s: a regression linear in the weekend fraction recovers a chord, log premium is strongly convex in that fraction away from the money, and on a simulated book generated by the pricer itself the same gap appears with nothing to explain it. `tests/test_price_observed.py` pins that, which is why the comparison quoted above is observed against reconstructed rather than observed against a linearization.

8. What this note does not establish

- **The counterfactual leg is modelled and always will be.** Section 7 puts the base on observed money and shows the effect measures the same either way, but “the same contract with no weekend in it” is something the tape supplies only across contracts, never for one contract against itself. It is a cross-sectional comparison, not a forecast: switching the clock off book-wide would move the level of the surface too, and nothing here estimates that.
- **The dollar totals cover the filtered short-dated sample only** — trades between 0.25 and 14 days with usable volatility and delta between 0.02 and 0.98. They are not Deribit’s option revenue and they are not annual figures; volume grew by orders of magnitude across the sample.
- **No standard errors on the price effects.** Each is a deterministic function of an estimated ratio, so its uncertainty is that ratio’s uncertainty, carried through a nonlinear map. The gap figures in §4 above inherit the paper’s own finding that no single asset’s gap is individually significant.

Reproduction

```
python scripts/weekend_price_impact.py
python scripts/weekend_price_observed.py
```

The first reads the headline pricing table, the smile-by-moneyness table, the vintage trajectory and the cached smile samples in `data/panels/`, and writes `p1-p8` and `p_f1` in about 70 seconds. The second writes `o1-o5` and takes around twenty minutes, most of it the Bitcoin regressions.

The smile-sample cache now carries the traded premium, the trade-time forward and the strike alongside the volatility; a cache written before those columns existed is rebuilt from the tape automatically, with a warning, the first time any of these scripts loads it.

Tests: `tests/test_price_impact.py` (12) pins the identities — the effect vanishes when $w = 0$ or $R = 1$, the repricing returns the planted weekday volatility, the dollar effect is invariant to which

leg of a strike is priced while the percentage is not, and the first-order vega approximation agrees with the exact repricing only where it is entitled to. `tests/test_price_observed.py` (10) builds a book whose premia were generated by a known clock and checks that the reconstruction returns exact, that the inverse and linear premium conventions are handled apart, that the observed-versus-reconstructed difference is zero there and non-zero when a real divergence is planted, and that the local-derivative comparison fails in the wings on the pricer's own prices — the reason it is reported and not relied on.